```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC

from sklearn.metrics import
accuracy_score,classification_report,confusion_matrix,f1_score

df=pd.read_csv("breast_cancer_data.csv")

df.head()
```

```
         id diagnosis  radius_mean  texture_mean  perimeter_mean
area_mean  \
0    842302         M        17.99         10.38          122.80
1001.0
1    842517         M        20.57         17.77          132.90
1326.0
2  84300903         M        19.69         21.25          130.00
1203.0
3  84348301         M        11.42         20.38           77.58
386.1
4  84358402         M        20.29         14.34          135.10
1297.0

   smoothness_mean  compactness_mean  concavity_mean  concave
points_mean  \
0          0.11840           0.27760          0.3001
0.14710
1          0.08474           0.07864          0.0869
0.07017
2          0.10960           0.15990          0.1974
0.12790
3          0.14250           0.28390          0.2414
0.10520
4          0.10030           0.13280          0.1980
0.10430

   ...  texture_worst  perimeter_worst  area_worst
smoothness_worst  \
0  ...          17.33           184.60      2019.0            0.1622
```

```
1  ...          23.41           158.80      1956.0            0.1238

2  ...          25.53           152.50      1709.0            0.1444

3  ...          26.50            98.87       567.7            0.2098

4  ...          16.67           152.20      1575.0            0.1374


   compactness_worst  concavity_worst  concave points_worst
symmetry_worst  \
0             0.6656           0.7119                0.2654
0.4601
1             0.1866           0.2416                0.1860
0.2750
2             0.4245           0.4504                0.2430
0.3613
3             0.8663           0.6869                0.2575
0.6638
4             0.2050           0.4000                0.1625
0.2364

   fractal_dimension_worst  Unnamed: 32
0                  0.11890          NaN
1                  0.08902          NaN
2                  0.08758          NaN
3                  0.17300          NaN
4                  0.07678          NaN

[5 rows x 33 columns]

df.shape

(569, 33)

df.columns

Index(['id', 'diagnosis', 'radius_mean', 'texture_mean',
'perimeter_mean',
       'area_mean', 'smoothness_mean', 'compactness_mean',
'concavity_mean',
       'concave points_mean', 'symmetry_mean',
'fractal_dimension_mean',
       'radius_se', 'texture_se', 'perimeter_se', 'area_se',
'smoothness_se',
       'compactness_se', 'concavity_se', 'concave points_se',
'symmetry_se',
       'fractal_dimension_se', 'radius_worst', 'texture_worst',
       'perimeter_worst', 'area_worst', 'smoothness_worst',
       'compactness_worst', 'concavity_worst', 'concave points_worst',
```

```
       'symmetry_worst', 'fractal_dimension_worst', 'Unnamed: 32'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 33 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   id                       569 non-null    int64  
 1   diagnosis                569 non-null    object 
 2   radius_mean              569 non-null    float64
 3   texture_mean             569 non-null    float64
 4   perimeter_mean           569 non-null    float64
 5   area_mean                569 non-null    float64
 6   smoothness_mean          569 non-null    float64
 7   compactness_mean         569 non-null    float64
 8   concavity_mean           569 non-null    float64
 9   concave points_mean      569 non-null    float64
 10  symmetry_mean            569 non-null    float64
 11  fractal_dimension_mean   569 non-null    float64
 12  radius_se                569 non-null    float64
 13  texture_se               569 non-null    float64
 14  perimeter_se             569 non-null    float64
 15  area_se                  569 non-null    float64
 16  smoothness_se            569 non-null    float64
 17  compactness_se           569 non-null    float64
 18  concavity_se             569 non-null    float64
 19  concave points_se        569 non-null    float64
 20  symmetry_se              569 non-null    float64
 21  fractal_dimension_se     569 non-null    float64
 22  radius_worst             569 non-null    float64
 23  texture_worst            569 non-null    float64
 24  perimeter_worst          569 non-null    float64
 25  area_worst               569 non-null    float64
 26  smoothness_worst         569 non-null    float64
 27  compactness_worst        569 non-null    float64
 28  concavity_worst          569 non-null    float64
 29  concave points_worst     569 non-null    float64
 30  symmetry_worst           569 non-null    float64
 31  fractal_dimension_worst  569 non-null    float64
 32  Unnamed: 32              0 non-null      float64
dtypes: float64(31), int64(1), object(1)
memory usage: 146.8+ KB
```

```
df.describe()
```

```
                 id  radius_mean  texture_mean  perimeter_mean
area_mean  \
```

```
count  5.690000e+02   569.000000    569.000000      569.000000
569.000000
mean   3.037183e+07    14.127292     19.289649       91.969033
654.889104
std    1.250206e+08     3.524049      4.301036       24.298981
351.914129
min    8.670000e+03     6.981000      9.710000       43.790000
143.500000
25%    8.692180e+05    11.700000     16.170000       75.170000
420.300000
50%    9.060240e+05    13.370000     18.840000       86.240000
551.100000
75%    8.813129e+06    15.780000     21.800000      104.100000
782.700000
max    9.113205e+08    28.110000     39.280000      188.500000
2501.000000

       smoothness_mean  compactness_mean  concavity_mean  concave
points_mean  \
count       569.000000        569.000000      569.000000
569.000000
mean          0.096360          0.104341        0.088799
0.048919
std           0.014064          0.052813        0.079720
0.038803
min           0.052630          0.019380        0.000000
0.000000
25%           0.086370          0.064920        0.029560
0.020310
50%           0.095870          0.092630        0.061540
0.033500
75%           0.105300          0.130400        0.130700
0.074000
max           0.163400          0.345400        0.426800
0.201200

       symmetry_mean  ...  texture_worst  perimeter_worst   area_worst
\
count     569.000000  ...     569.000000       569.000000   569.000000

mean        0.181162  ...      25.677223       107.261213   880.583128

std         0.027414  ...       6.146258        33.602542   569.356993

min         0.106000  ...      12.020000        50.410000   185.200000

25%         0.161900  ...      21.080000        84.110000   515.300000

50%         0.179200  ...      25.410000        97.660000   686.500000
```

```
75%          0.195700  ...      29.720000       125.400000  1084.000000

max          0.304000  ...      49.540000       251.200000  4254.000000


       smoothness_worst  compactness_worst  concavity_worst  \
count        569.000000         569.000000       569.000000
mean           0.132369           0.254265         0.272188
std            0.022832           0.157336         0.208624
min            0.071170           0.027290         0.000000
25%            0.116600           0.147200         0.114500
50%            0.131300           0.211900         0.226700
75%            0.146000           0.339100         0.382900
max            0.222600           1.058000         1.252000

       concave points_worst  symmetry_worst
fractal_dimension_worst  \
count             569.000000      569.000000               569.000000

mean                0.114606        0.290076                 0.083946

std                 0.065732        0.061867                 0.018061

min                 0.000000        0.156500                 0.055040

25%                 0.064930        0.250400                 0.071460

50%                 0.099930        0.282200                 0.080040

75%                 0.161400        0.317900                 0.092080

max                 0.291000        0.663800                 0.207500


       Unnamed: 32
count          0.0
mean           NaN
std            NaN
min            NaN
25%            NaN
50%            NaN
75%            NaN
max            NaN

[8 rows x 32 columns]
```

```
df.isnull().sum()
```

```
id                         0
diagnosis                  0
radius_mean                0
```

```
texture_mean                0
perimeter_mean              0
area_mean                   0
smoothness_mean             0
compactness_mean            0
concavity_mean              0
concave points_mean         0
symmetry_mean               0
fractal_dimension_mean      0
radius_se                   0
texture_se                  0
perimeter_se                0
area_se                     0
smoothness_se               0
compactness_se              0
concavity_se                0
concave points_se           0
symmetry_se                 0
fractal_dimension_se        0
radius_worst                0
texture_worst               0
perimeter_worst             0
area_worst                  0
smoothness_worst            0
compactness_worst           0
concavity_worst             0
concave points_worst        0
symmetry_worst              0
fractal_dimension_worst     0
Unnamed: 32               569
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df=df.drop_duplicates()
```

```
if 'id' in df.columns:
    df.drop('id',axis=1,inplace=True)

if 'Unnamed: 32' in df.columns:
    df.drop('Unnamed: 32',axis=1,inplace=True)

df.isnull().sum()
```

```
diagnosis                 0
radius_mean               0
texture_mean              0
perimeter_mean            0
area_mean                 0
```

```
smoothness_mean             0
compactness_mean            0
concavity_mean              0
concave points_mean         0
symmetry_mean               0
fractal_dimension_mean      0
radius_se                   0
texture_se                  0
perimeter_se                0
area_se                     0
smoothness_se               0
compactness_se              0
concavity_se                0
concave points_se           0
symmetry_se                 0
fractal_dimension_se        0
radius_worst                0
texture_worst               0
perimeter_worst             0
area_worst                  0
smoothness_worst            0
compactness_worst           0
concavity_worst             0
concave points_worst        0
symmetry_worst              0
fractal_dimension_worst     0
dtype: int64
```

```
df.fillna(df.median(numeric_only=True),inplace=True)

le=LabelEncoder()

df['diagnosis']=le.fit_transform(df['diagnosis'])

df.head()
```

```
   diagnosis  radius_mean  texture_mean  perimeter_mean  area_mean  \
0          1        17.99         10.38          122.80     1001.0
1          1        20.57         17.77          132.90     1326.0
2          1        19.69         21.25          130.00     1203.0
3          1        11.42         20.38           77.58      386.1
4          1        20.29         14.34          135.10     1297.0

   smoothness_mean  compactness_mean  concavity_mean  concave
points_mean  \
0          0.11840           0.27760          0.3001
0.14710
1          0.08474           0.07864          0.0869
0.07017
2          0.10960           0.15990          0.1974
```

```
0.12790
3          0.14250           0.28390          0.2414
0.10520
4          0.10030           0.13280          0.1980
0.10430

   symmetry_mean  ...  radius_worst  texture_worst  perimeter_worst  \
0         0.2419  ...         25.38          17.33           184.60
1         0.1812  ...         24.99          23.41           158.80
2         0.2069  ...         23.57          25.53           152.50
3         0.2597  ...         14.91          26.50            98.87
4         0.1809  ...         22.54          16.67           152.20

   area_worst  smoothness_worst  compactness_worst  concavity_worst  \
0      2019.0            0.1622             0.6656           0.7119
1      1956.0            0.1238             0.1866           0.2416
2      1709.0            0.1444             0.4245           0.4504
3       567.7            0.2098             0.8663           0.6869
4      1575.0            0.1374             0.2050           0.4000

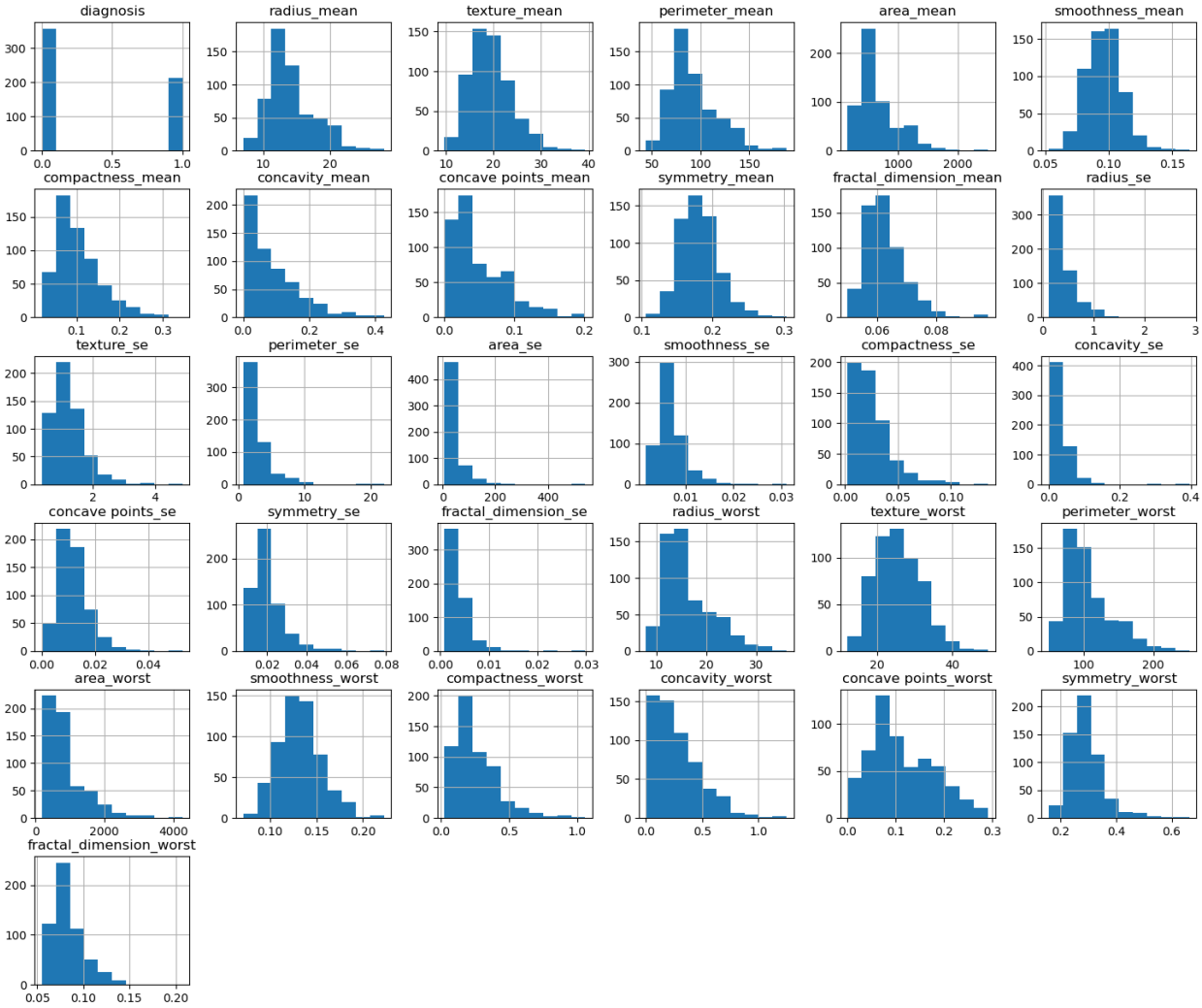
   concave points_worst  symmetry_worst  fractal_dimension_worst
0                0.2654          0.4601                  0.11890
1                0.1860          0.2750                  0.08902
2                0.2430          0.3613                  0.08758
3                0.2575          0.6638                  0.17300
4                0.1625          0.2364                  0.07678

[5 rows x 31 columns]
```

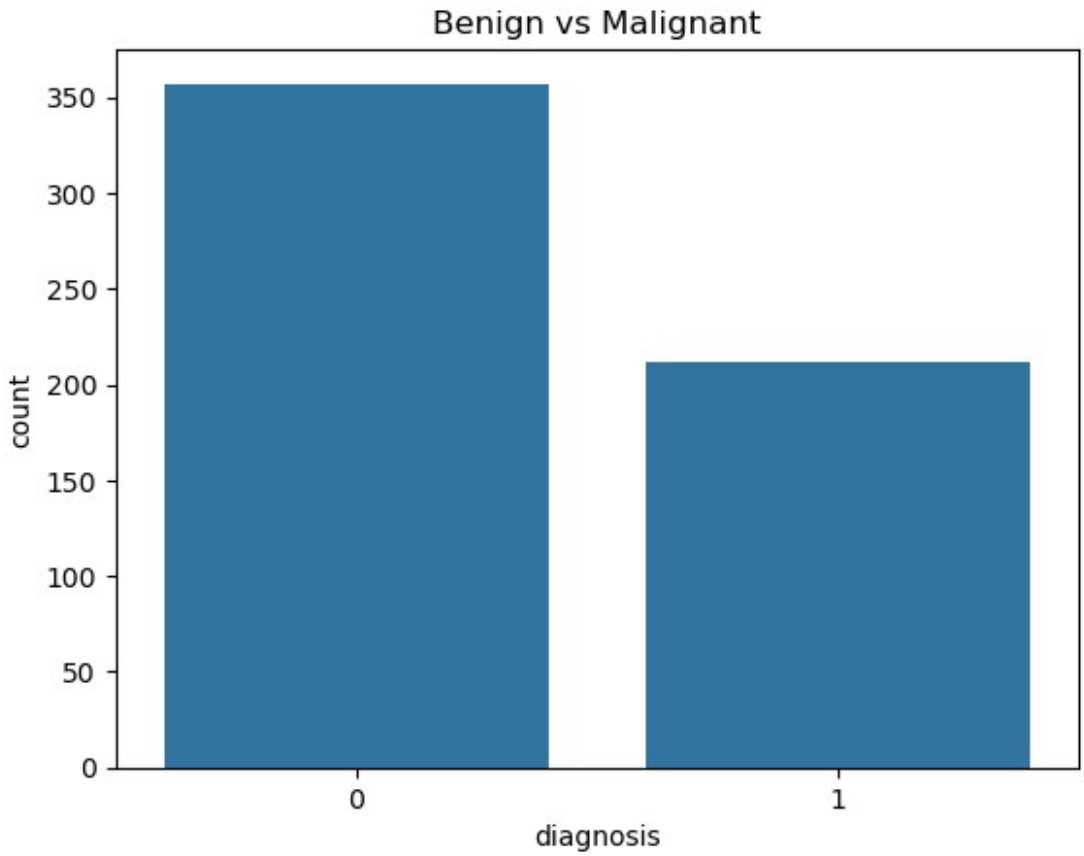
```
df['diagnosis'].value_counts()
```

```
diagnosis
0    357
1    212
Name: count, dtype: int64
```

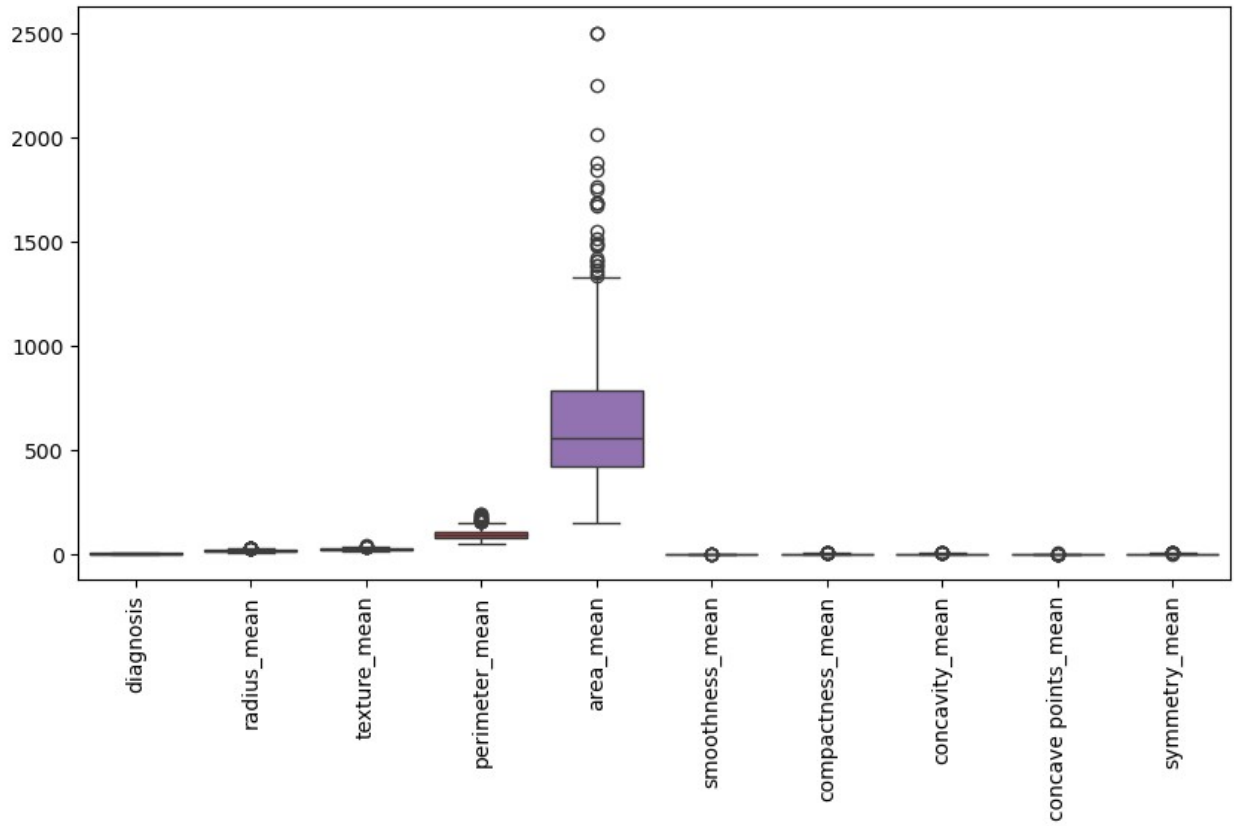
```
df.hist(figsize=(18,15))
plt.show()
```



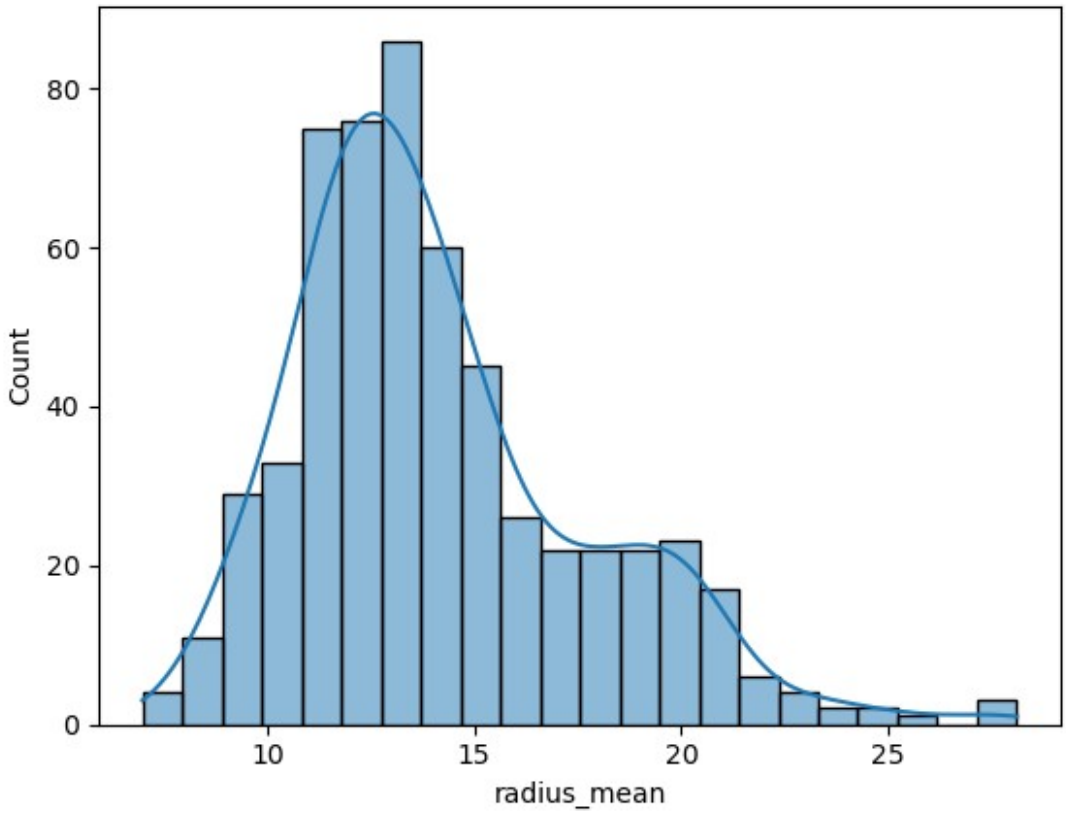

```
sns.countplot(x='diagnosis',data=df)
plt.title("Benign vs Malignant")
plt.show()
```



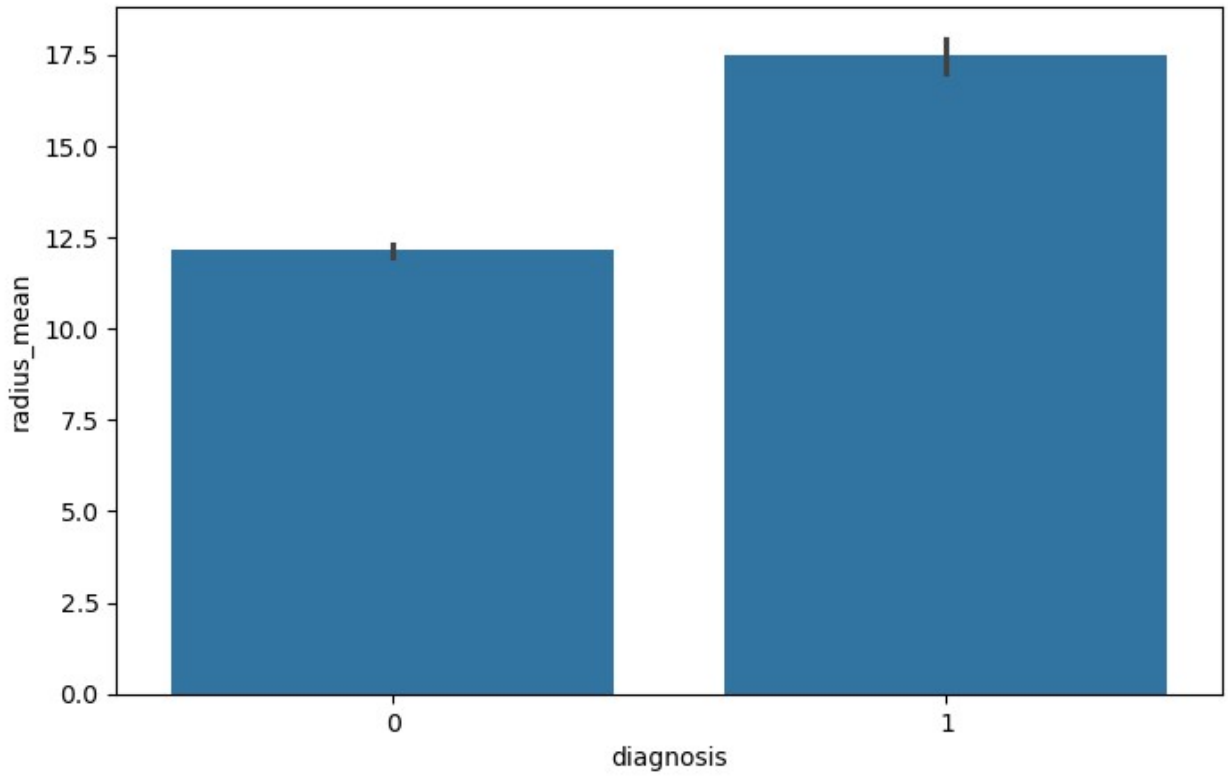

```
plt.figure(figsize=(10,5))
sns.boxplot(data=df.iloc[:,:10])
plt.xticks(rotation=90)
plt.show()
```



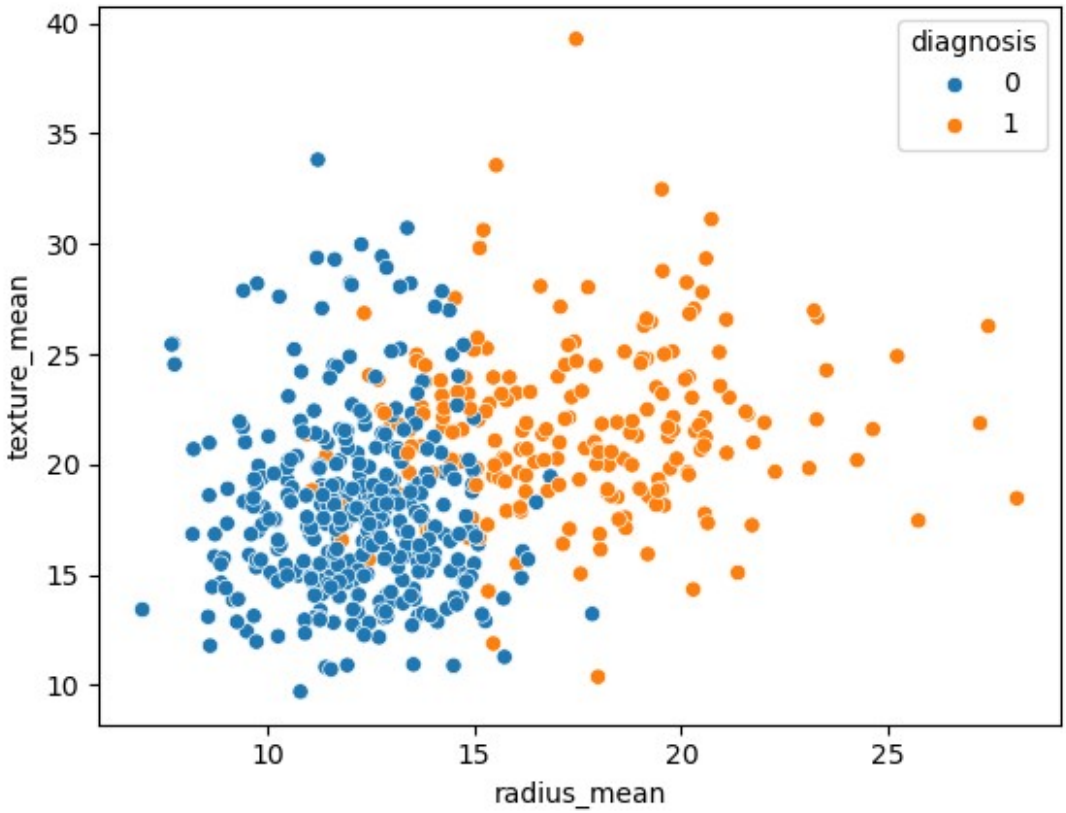

```
sns.histplot(df['radius_mean'],kde=True)
plt.show()
```



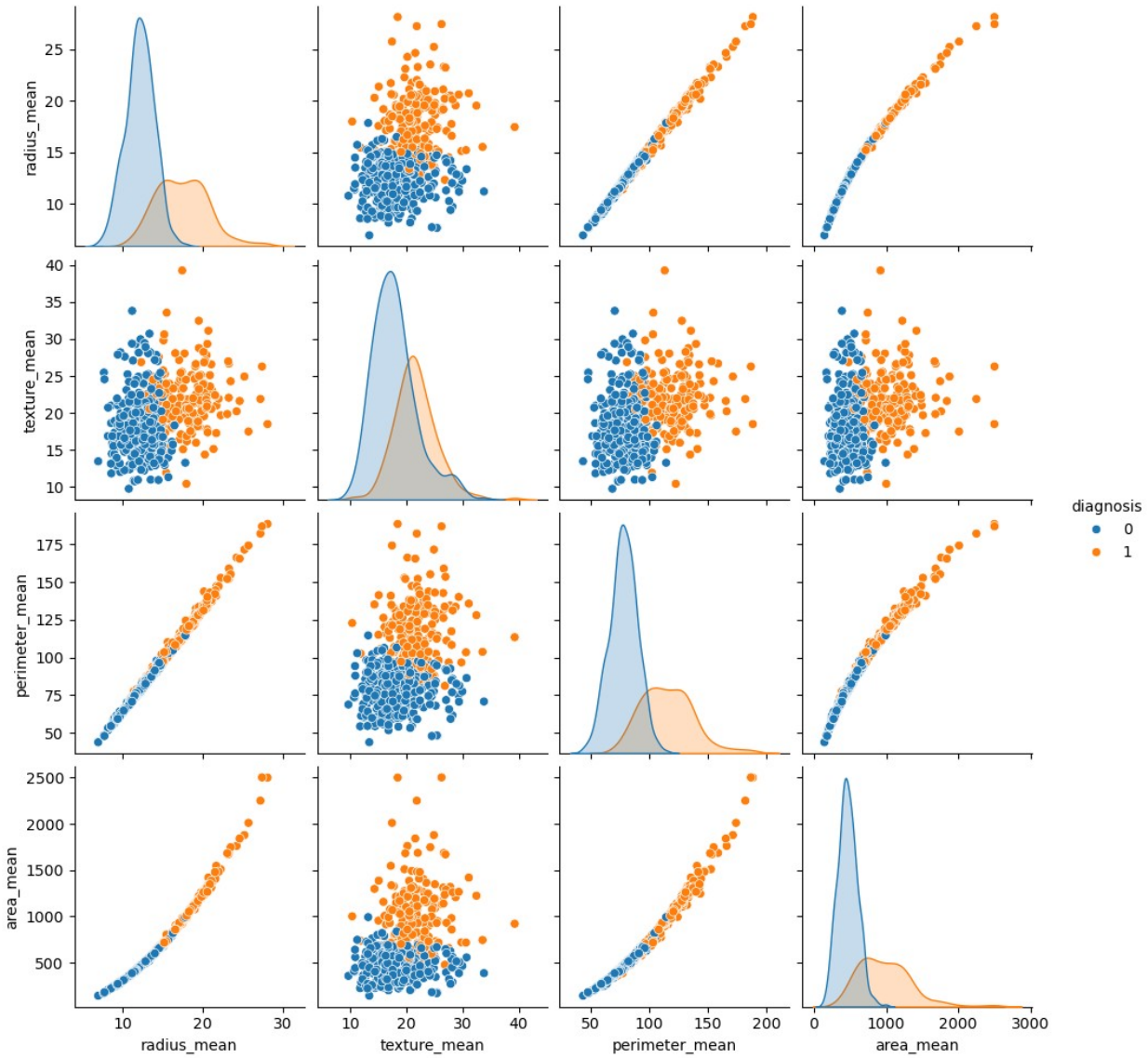

```
plt.figure(figsize=(8,5))
sns.barplot(x='diagnosis',y='radius_mean',data=df)
plt.show()
```



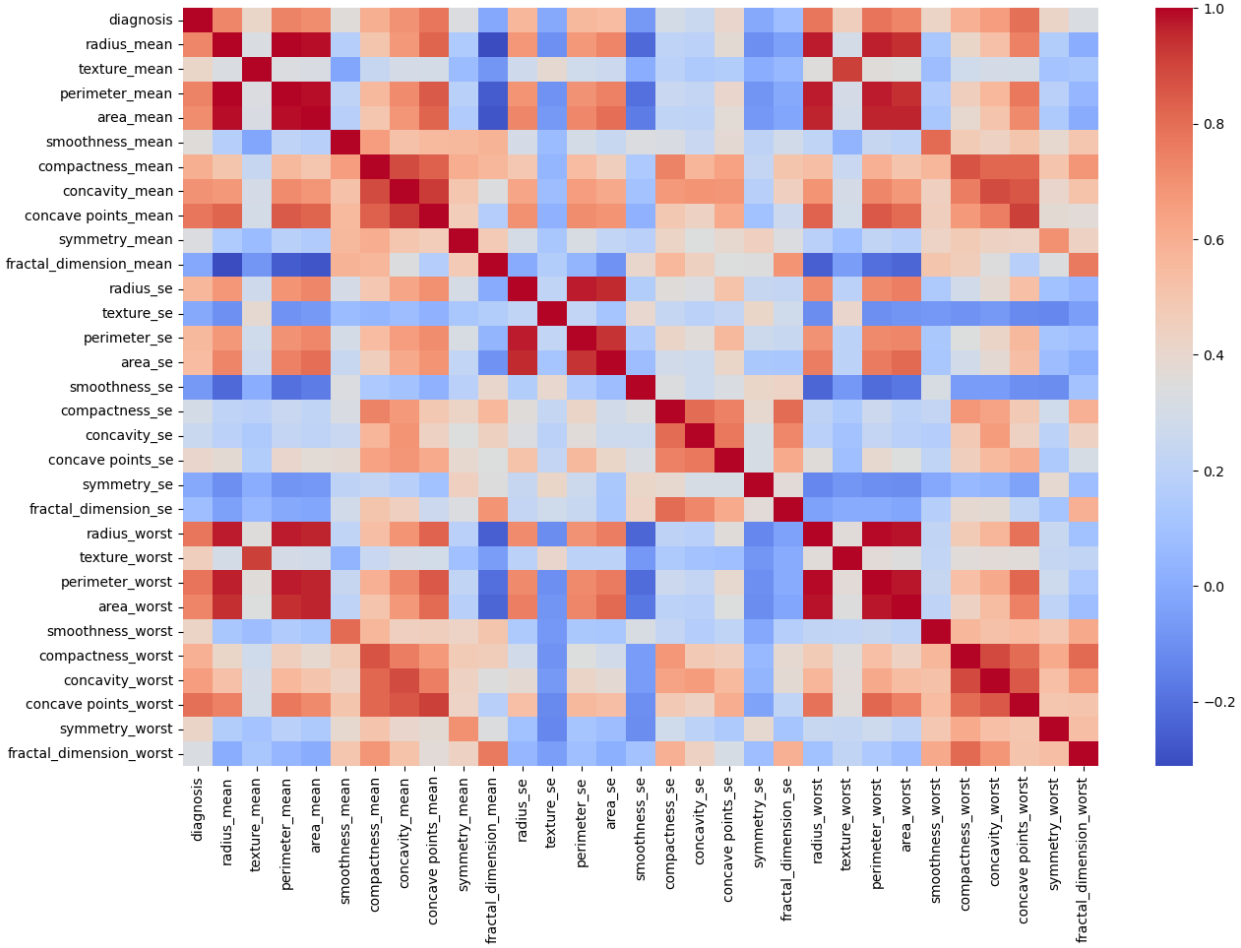

```
sns.scatterplot(x='radius_mean',
                y='texture_mean',
                hue='diagnosis',
                data=df)
plt.show()
```




```
sns.pairplot(df[['radius_mean',
                 'texture_mean',
                 'perimeter_mean',
                 'area_mean',
                 'diagnosis']],
             hue='diagnosis')
plt.show()
```




```
plt.figure(figsize=(15,10))
sns.heatmap(df.corr(),cmap='coolwarm')
plt.show()
```




```
X=df.drop('diagnosis',axis=1)

y=df['diagnosis']

scaler=StandardScaler()

X=scaler.fit_transform(X)

X_train,X_test,y_train,y_test=train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

lr=LogisticRegression()

lr.fit(X_train,y_train)

pred_lr=lr.predict(X_test)
```

```
print("Accuracy:",accuracy_score(y_test,pred_lr))
```

```
Accuracy: 0.9649122807017544
```

```
dt=DecisionTreeClassifier(random_state=42)
```

```
dt.fit(X_train,y_train)
```

```
pred_dt=dt.predict(X_test)
```

```
print("Accuracy:",accuracy_score(y_test,pred_dt))
```

```
Accuracy: 0.9298245614035088
```

```
rf=RandomForestClassifier(random_state=42)
```

```
rf.fit(X_train,y_train)
```

```
pred_rf=rf.predict(X_test)
```

```
print("Accuracy:",accuracy_score(y_test,pred_rf))
```

```
Accuracy: 0.9736842105263158
```

```
svm=SVC()
```

```
svm.fit(X_train,y_train)
```

```
pred_svm=svm.predict(X_test)
```

```
print("Accuracy:",accuracy_score(y_test,pred_svm))
```

```
Accuracy: 0.9736842105263158
```

```
results=pd.DataFrame({

'Model':['Logistic Regression',
         'Decision Tree',
         'Random Forest',
         'Support Vector Machine'],

'Accuracy':[accuracy_score(y_test,pred_lr),
            accuracy_score(y_test,pred_dt),
            accuracy_score(y_test,pred_rf),
            accuracy_score(y_test,pred_svm)]
})
```

```
results
```

```
                    Model  Accuracy
0     Logistic Regression  0.964912
```

```
1           Decision Tree  0.929825
2           Random Forest  0.973684
3  Support Vector Machine  0.973684
```

```
print(classification_report(y_test,pred_rf))
```

```
              precision    recall  f1-score   support

           0       0.96      1.00      0.98        72
           1       1.00      0.93      0.96        42

    accuracy                           0.97       114
   macro avg       0.98      0.96      0.97       114
weighted avg       0.97      0.97      0.97       114
```

```
print("F1 Score:",f1_score(y_test,pred_rf))
```

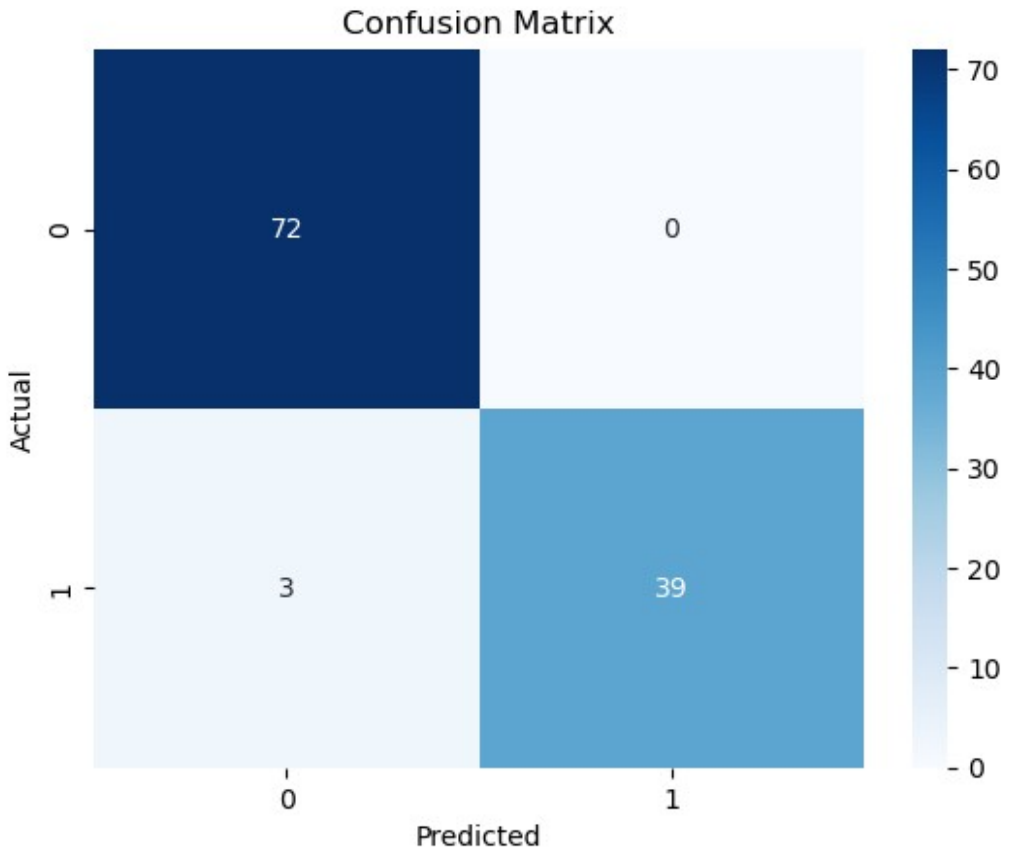
```
F1 Score: 0.9629629629629629
```

```
cm=confusion_matrix(y_test,pred_rf)

sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues')

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```




```
print("Random Forest Accuracy:",accuracy_score(y_test,pred_rf))
```

```
Random Forest Accuracy: 0.9736842105263158
```